USER GUIDE

PharmaCare + PharmaIQ User Guide

How to navigate the system, what to ask,
and what to expect back

Project: PharmaCare + PharmaIQ (Flask · SQLite · pandas · hybrid RAG · optional LLM)
Document date: 15 October 2026 · Based on the code and demo data in PharmaCare-PharmaIQ.zip

Contents

1. What this project is . . . 3

What you can do . . . 3

2. Getting started . . . 4

2.1 Install and run . . . 4

2.2 Log in . . . 4

2.3 Your first five minutes . . . 4

3. Finding your way around . . . 5

Role summary . . . 5

4. The PharmaCare pages . . . 6

4.1 Dashboard . . . 6

4.2 Medicines . . . 6

4.3 Sales . . . 7

4.4 Alerts . . . 7

4.5 Barcode . . . 7

4.6 Reports and Export (admin, pharmacist) . . . 7

4.7 Users, Activity Logs, Profile . . . 7

5. Ask PharmaIQ — the chat assistant . . . 8

5.1 Follow-up questions . . . 9

6. Prompt library with real outputs . . . 10

6.1 Expiry and stock . . . 10

6.2 Value and sales . . . 10

6.3 Analytics (forecast, anomalies and reorder are admin / pharmacist) . . . 11

6.4 Documents (leaflets and SOPs) . . . 12

6.5 When PharmaIQ declines or says “I don’t know” . . . 12

6.6 More phrasings that were tested and work . . . 13

6.7 Phrasings that do not work well (yet) . . . 13

7. Knowledge Base — documents PharmaIQ can cite . . . 14

7.1 Upload your own document . . . 14

7.2 How document answers look . . . 15

8. AI Insights (admin and pharmacist) . . . 15

8.1 Demand forecast . . . 16

8.2 Anomalies and data quality . . . 16

8.3 Reorder suggestions . . . 16

9. Admin tools: Audit Trail and Evaluation . . . 17

9.1 AI Audit Trail . . . 17

9.2 AI Evaluation 17
10. What PharmaIQ will and will not do 17
Staff see only their own sales 18
11. Tips for getting good answers 18
12. Troubleshooting and FAQ 19
Frequently asked questions 19
13. Glossary 20

1. What this project is

PharmaCare is a web application for running a pharmacy: it tracks medicines (stock, batches, expiry, price, barcode), records sales, raises alerts for expired / expiring / low-stock items, produces reports and Excel exports, and has three user roles (admin, pharmacist, staff).

PharmaIQ is an AI assistant built into PharmaCare. Instead of clicking through pages, you simply *ask* a question in plain English — "which medicines expire in 30 days?", "forecast demand for paracetamol", "how should insulin be stored?" — and PharmaIQ answers from your real data and your uploaded documents.

PharmaIQ was designed around one rule: **never make up an answer**. Every response states where it came from, shows the real rows or document passages behind it, and says "I won't guess" when it cannot find support.

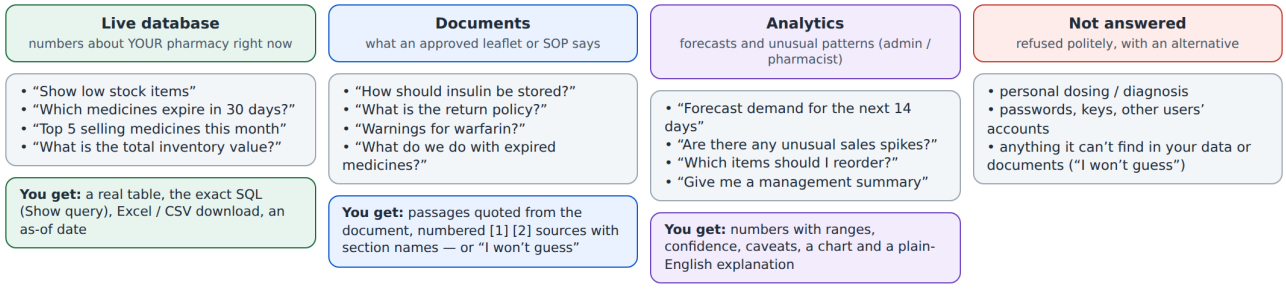


Figure 1 – The four kinds of answer PharmaIQ gives, and what you receive with each.

What you can do

- **Run the pharmacy** (core PharmaCare pages): dashboard, medicines, sales, alerts, barcode lookup, reports, Excel export, users and activity log.
- **Ask questions about live data**: stock, expiry, low stock, inventory value, sales by period / category / seller.
- **Ask questions about documents**: medicine leaflets and your own SOPs, with numbered citations.
- **Get analytics** (admin / pharmacist): demand forecast with a confidence range, unusual-sales and data-quality checks, reorder suggestions, a one-page management summary.
- **Download results** as Excel or CSV straight from an answer; see the exact SQL behind any table.
- **Stay safe**: personal dosing / diagnosis questions and credential requests are politely refused; every question is logged for admins to review.

> **Good to know**
> PharmaIQ runs **fully offline by default**: no internet, no API key and no external AI service are required. An optional large language model (LLM) can be switched on by an administrator to improve wording and understanding — see the Technical Guide.

2. Getting started

2.1 Install and run

You need **Python 3.9 or newer**. From the unzipped project folder:

```
cd PharmaCare-PharmaIQ
python -m venv venv
source venv/bin/activate            # Windows:  venv\Scripts\activate
pip install -r requirements.txt
python seed_ai_demo.py              # optional: loads 30 demo medicines, ~1,700 sales, 6 leaflets + 6 SOPs
python app.py                       # then open http://127.0.0.1:5000
```

The first start creates the database (`instance/pharmacy.db`). The optional `seed_ai_demo.py` step is strongly recommended for a first visit: without data and documents, most questions have nothing to answer from.

2.2 Log in

Role	Username	Password	What this role can do
Admin	admin	admin123	Everything, including users, activity logs, AI Audit Trail and AI Evaluation
Pharmacist	pharmacist	pharma123	Medicines, sales, alerts, reports, Excel export, AI Insights, document upload
Staff	staff	staff123	Dashboard, medicines, sales, alerts, barcode, Ask PharmaIQ and view Knowledge Base

> **Security**
> These are demo credentials. Change every password and set a strong SECRET_KEY before using the system with real data.

2.3 Your first five minutes

- Log in as **admin** and look at the **Dashboard** — it shows what needs attention today (expired, expiring, low stock).
- Open **Ask PharmaIQ** in the sidebar and click one of the suggestion chips, or type *“Show low stock items”*.
- Click **Show query** under the table to see the exact SQL that produced the numbers.
- Click a follow-up chip (for example *“Now show only tablets”*) — PharmaIQ remembers the previous question.
- Ask a document question: *“How should insulin be stored?”* — note the numbered sources.
- Open **AI Insights** and press **Forecast** to see a demand forecast with its uncertainty band.

3. Finding your way around

After login you see a top bar (notifications, your name and role, profile / logout) and a left sidebar. What appears in the sidebar depends on your role.

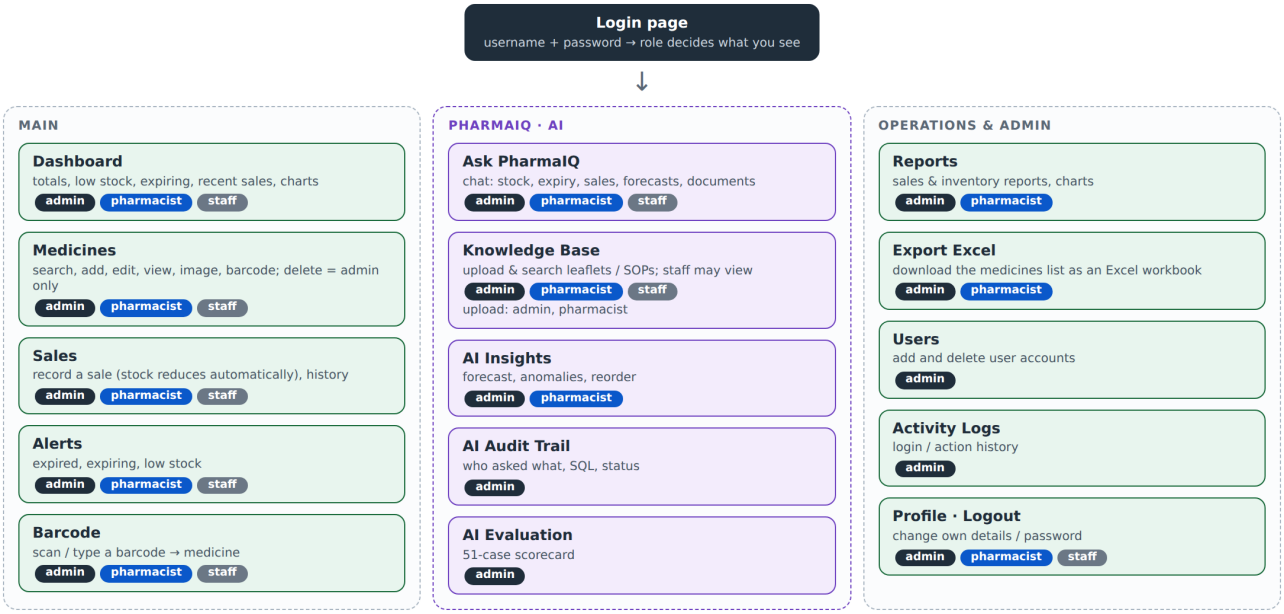


Figure 2 – Screen map. Coloured chips show which roles can open each page.

Role summary

Page	Admin	Pharmacist	Staff
Dashboard, Medicines, Sales, Alerts, Barcode	Yes	Yes	Yes
Ask PharmaIQ	Yes	Yes	Yes (sales questions limited to own sales)
Knowledge Base (view / search)	Yes	Yes	Yes (documents marked “All staff”)
Knowledge Base (upload / delete)	Upload + delete	Upload only	No
AI Insights (forecast, anomalies, reorder)	Yes	Yes	No
Reports, Export Excel	Yes	Yes	No
AI Audit Trail, AI Evaluation	Yes	No	No
Users, Activity Logs	Yes	No	No

4. The PharmaCare pages

These pages are the original pharmacy application. PharmaIQ does not change them — it only *reads* the same data. (Pages below are shown as illustrative renderings built from the real page layouts and the demo data.)

4.1 Dashboard

The landing page. Top row: **Total Medicines**, **Total Stock Units**, **Inventory Value** and **Total Sales**. Second row: three alert cards — **Expired!**, **Expiring Soon** (within 30 days) and **Low Stock** (below the threshold, default 10 units) — each linking to the Alerts page. Below: a **Sales — Last 7 Days** line chart, a **Stock Overview** doughnut (Normal / Low / Expiring), **Recent Medicines** and **Recent Activity**.

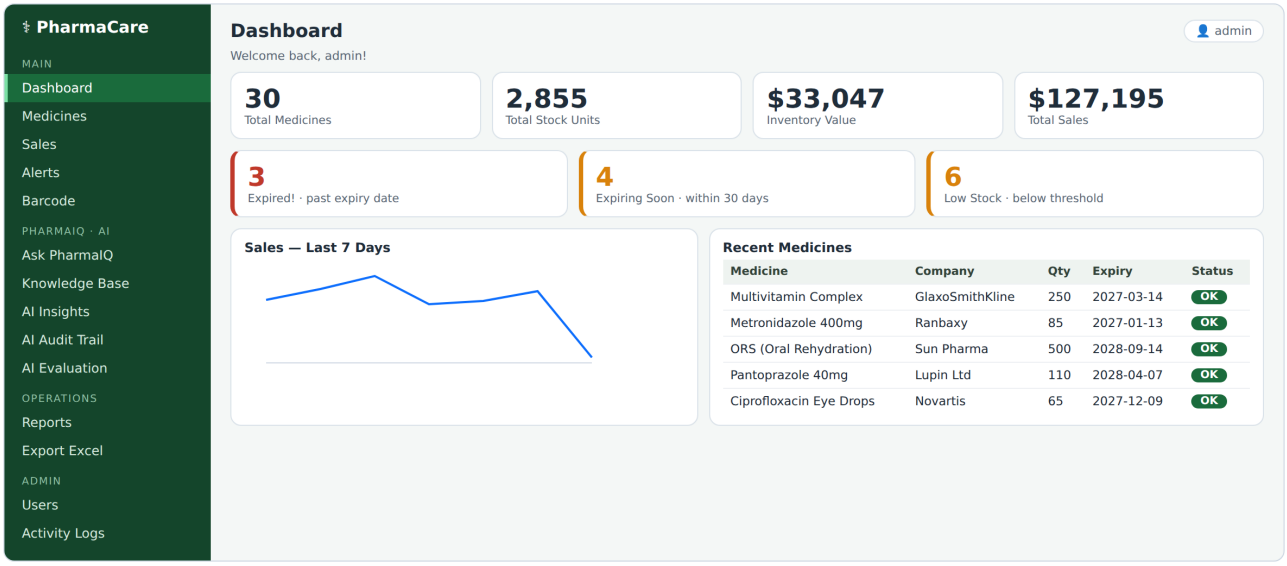


Figure 3 – Dashboard (illustrative rendering using the demo data).

4.2 Medicines

The full inventory list. Use the search box (name, company, type or barcode). Rows are colour-flagged: red = expired, amber = expiring soon / low stock. The **Stock** badge is red / amber / green by quantity and **Status** shows Expired, Expiring, Low Stock or OK.

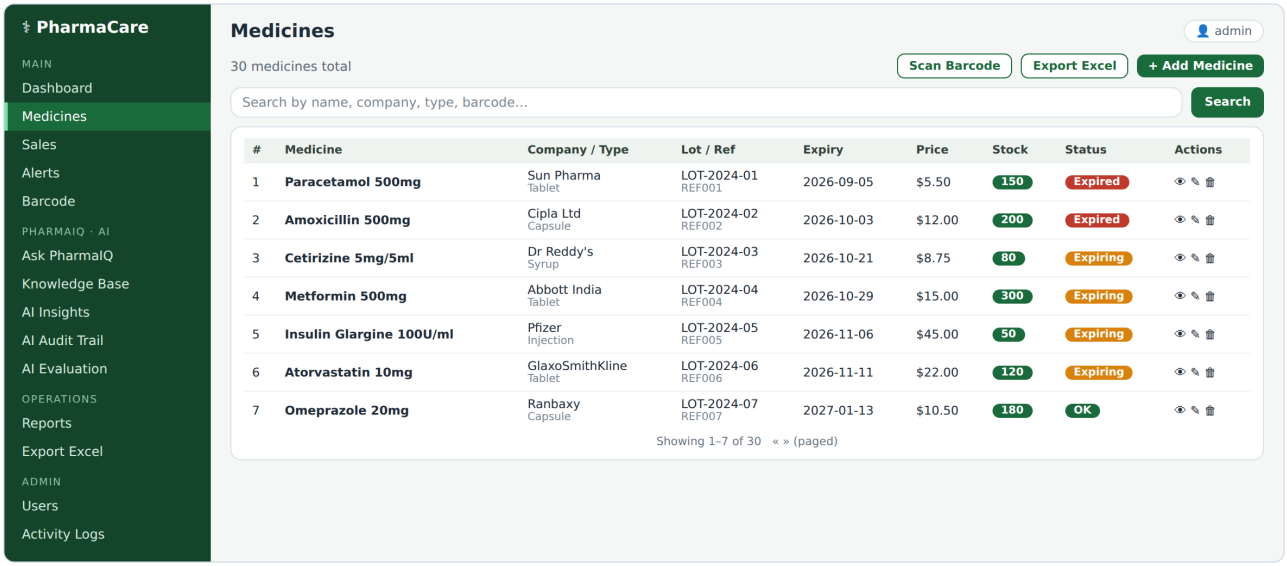


Figure 4 – Medicines list.

- **Add Medicine** — opens a form: reference no., company, type (category), name, lot no., issue and expiry dates, uses, side effects, precautions, dosage, price, quantity, barcode and an optional image.
- **View** (eye icon) shows the full detail page; **Edit** (pencil) opens the same form pre-filled; **Delete** (bin) removes a medicine and is **admin only**.
- **Scan Barcode** jumps to the barcode page; **Export Excel** downloads the medicine list as a workbook.

4.3 Sales

Lists every sale (medicine, quantity, unit price, total, who sold it, date), paged. Press **New Sale**: choose a medicine (its current stock is shown), enter the quantity — the unit price fills in and the total updates — then **Record Sale**. Stock is reduced automatically, and the sale is attributed to your username (this is what lets staff later see “my sales” in PharmaIQ).

4.4 Alerts

Three sections that need action: **Expired Medicines** (“immediate action required”), **Expiring Within 30 Days** (with days left) and **Low Stock Items**. The thresholds are 30 days and 10 units by default and can be changed by an administrator (see Technical Guide, Configuration).

4.5 Barcode

Two ways to look up a medicine: **Camera** (press Start Scanner and show the barcode) or **Manual Entry** (type or paste the number and press Lookup). Matches show the medicine; unknown codes show “Barcode not found in system”. **Recent Scans** keeps a clearable history.

4.6 Reports and Export (admin, pharmacist)

Reports & Analytics shows the number of categories, total inventory value, low-stock and expiring counts, a stock-by-type chart, an Inventory-by-Category table (type, count, quantity, value), the Top 10 Low Stock medicines and everything expiring within 30 days. **Export to Excel** downloads a workbook.

4.7 Users, Activity Logs, Profile

- **Users** (admin): list accounts and roles; **Add User** (username, password, role, name, email); delete an account.
- **Activity Logs** (admin): who did what and when (logins, edits, exports…).
- **Profile** (everyone): update your name and email; change your password by entering the current one first.

5. Ask PharmaIQ — the chat assistant

Open **Ask PharmaIQ** in the sidebar. Type a question in everyday language and press **Send**. There are no commands to learn. The page also offers starter suggestion chips and a **New chat** button to clear the conversation context.

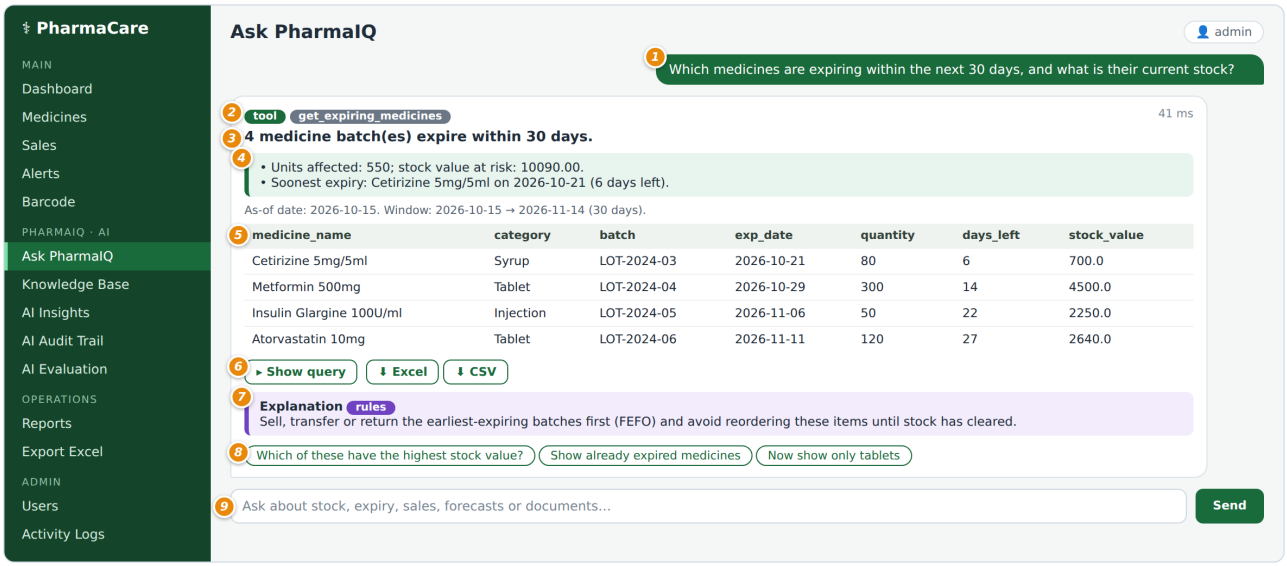


Figure 5 – Anatomy of an answer (real output for the question shown, demo data).

#	Part of the answer	What it tells you
1	Your question	Up to 600 characters. Questions are remembered within a chat so you can follow up.
2	Route badge	**tool** = a built-in report; **sql** = a custom read-only query built for your question; **documents** = answered from uploaded leaflets / SOPs; **refused** = out of scope; **help** = capabilities. Shows the tool name and response time.
3	Headline	A one-line answer computed from the data (for example, “4 medicine batch(es) expire within 30 days.”).
4	Facts (green)	Deterministic statements calculated by code from the real rows — totals, soonest expiry, stock value at risk. These are the numbers to trust.
5	Result table	The real database rows. An as-of date and any assumptions (period, thresholds) are printed just above it.
6	Show query / Excel / CSV	**Show query** reveals the exact SQL and parameters. **Excel** and **CSV** download the table (Excel for formatted use, CSV for other tools).
7	Explanation (purple)	Short plain-English guidance. A label says whether it was written by *rules* or by an optional *model*. If a model’s text contains a number that is not in the data, it is discarded automatically.
8	Follow-up chips	One click asks a sensible next question; the assistant keeps the earlier filters.
9	Input box	Type your next question. Charts appear automatically when a result is best shown visually (rankings, trends).

5.1 Follow-up questions

PharmaIQ keeps the *structure* of your previous query, not just the words. After “Show all products with fewer than 10 units” (5 rows) you can type **“Now show only tablets”** and get the same filter plus the new condition (2 rows). Verified follow-ups that work: *“Now show only tablets”, “Now show only injections”, “Now show only from Cipla”*. Start follow-ups with **“Now…”** and name a category or company. Follow-ups that only change the *sort order* (“Now sort by price”) are not understood yet — ask a new question instead, such as *“Show the 5 most expensive medicines”*. Press **New chat** to start fresh.

> **Reading tips**
> **If a document answer has no table:** that is normal — document answers show quoted passages and a Sources list instead of rows.
> **Two different sources are never mixed silently:** the route badge always tells you whether you are reading database results or document text.

6. Prompt library with real outputs

Every example below was run against the demo database on 15 Oct 2026 (as-of date 2026-10-15). Your numbers will differ with your own data. The coloured strip on the right of each question shows the route and tool that handled it.

6.1 Expiry and stock

Ask: “Which medicines are expiring within the next 30 days, and what is their current stock?” **tool · get_expiring_medicines**

4 medicine batch(es) expire within 30 days.
• Units affected: 550; stock value at risk: 10090.00.
• Soonest expiry: Cetirizine 5mg/5ml on 2026-10-21 (6 days left).

Table: **Medicines expiring within 30 days** — columns: medicine_name, category, batch, exp_date, quantity, days_left, stock_value — 4 row(s). First row: Cetirizine 5mg/5ml | Syrup | LOT-2024-03 | 2026-10-21 | 80 | 6 | 700.0

Ask: “Show already expired medicines” **tool · get_expired_medicines**

3 expired batch(es) still in the system, 357 units, value 3351.00.

Table: **Already expired medicines** — columns: medicine_name, category, batch, exp_date, quantity, days_overdue, stock_value — 3 row(s). First row: Paracetamol 500mg | Tablet | LOT-2024-01 | 2026-09-05 | 150 | 40 | 825.0

Ask: “Show low stock items” **tool · get_low_stock_items**

6 product(s) are at or below 10 units.

Table: **Low-stock items (≤ 10 units)** — columns: medicine_name, category, company, quantity, price, exp_date — 6 row(s). First row: Digoxin 0.25mg | Tablet | Dr Reddy's | 2 | 28.0 | 2027-01-13

Ask: “Show all products with fewer than 10 units” **sql · text_to_sql**

5 row(s) returned.

Table: **Medicines with units in stock below 10** — columns: medicine_name, category, company, quantity, price, exp_date — 5 row(s). First row: Digoxin 0.25mg | Tablet | Dr Reddy's | 2 | 28.0 | 2027-01-13

Then ask **“Now show only tablets”** — the same filter plus category, now 2 rows.

Ask: “Which medicines are priced above $20?” **sql · text_to_sql**

8 row(s) returned.

Table: **Medicines with price above 20** — columns: medicine_name, category, company, quantity, price, exp_date — 8 row(s). First row: Atorvastatin 10mg | Tablet | GlaxoSmithKline | 120 | 22.0 | 2026-11-11

Ask: “How many tablets do we have?” **sql · text_to_sql**

1 row(s) returned.

Table: **Medicine count** — columns: medicines — 1 row(s). First row: 13

6.2 Value and sales

Ask: “What is the total inventory value?” **tool · get_inventory_value**

Total inventory value: 33047.00 (price × quantity).

Table: **Inventory value** — columns: products, units, stock_value — 1 row(s). First row: 30 | 2855 | 33047.0

Ask: “Show the top 5 selling medicines this month” **tool · get_top_selling**

Best seller by units: Metformin 500mg (113 units, revenue 1695.00).

Table: **Top 5 selling medicines by units — this month (2026-10-01 → 2026-10-15)** — columns: medicine, units, revenue, transactions — 5 row(s). First row: Metformin 500mg | 113 | 1695.0 | 15

Ask: “Show sales by category this month” **tool · get_category_sales**

Total revenue in period: 20308.00 across 236 transaction(s).

• Top category: Tablet with revenue 8473.50 (42% of total).

Table: **Sales by category — this month (2026-10-01 → 2026-10-15)** — columns: category, transactions, units, revenue — 8 row(s). First row: Tablet | 100 | 488 | 8473.5

6.3 Analytics (forecast, anomalies and reorder are admin / pharmacist)

Ask: “Forecast demand for the next 14 days” **tool · forecast_demand**

Expected demand over the next 14 days: 1117 units (80% range 975–1260).

• Confidence: high (120 days of history, variability (CV) 0.29).

• Back-test (last 7 days): average error 25.5 units/day vs 25.8 for a naive average.

Table: **Forecast — next 14 days** — columns: date, forecast_units, lower_80, upper_80 — 14 row(s). First row: 2026-10-16 | 73.8 | 59.3 | 88.3

Ask: “Forecast demand for paracetamol” **tool · forecast_demand**

Expected demand over the next 14 days: 34 units (80% range 3–64).

• Confidence: medium (120 days of history, variability (CV) 0.93).

• Back-test (last 7 days): average error 3.6 units/day vs 3.6 for a naive average.

Table: **Forecast — next 14 days** — columns: date, forecast_units, lower_80, upper_80 — 14 row(s). First row: 2026-10-16 | 2.0 | 0.0 | 5.3

Notice the much wider range (3–64 units) — a single slow-moving product is far less predictable than the whole shop.

Ask: “Are there any unusual sales spikes?” **tool · detect_anomalies**

12 finding(s): 1× sale total mismatch, 7× sales spike, 4× sold after expiry.

Table: **Anomalies & data-quality findings** — columns: type, severity, entity, detail, date — 12 row(s). First row: sold_after_expiry | high | Amoxicillin 500mg (sale #15) | Sold on 2026-10-08 but expired 2026-10-03 |

Ask: “Which items should I reorder?” **tool · get_reorder_suggestions**

7 product(s) should be reviewed for reordering.

Table: **Reorder suggestions** — columns: medicine_name, category, quantity, units_sold_30d, avg_daily_sales, days_of_cover, suggested_order_qty — 7 row(s). First row: Warfarin 5mg | Tablet | 3 | 173 | 5.77 | 0.5 | 171

Ask: “Give me a management summary” **tool · get_management_summary**

Inventory: 30 products, 2855 units, value 33047.00.

• 6 product(s) are at or below 10 units.

• 4 medicine batch(es) expire within 30 days.

Table: **Inventory KPIs** — columns: products, units, stock_value — 1 row(s). First row: 30 | 2855 | 33047.0

The management summary is available to every role; forecasts, anomaly checks and reorder suggestions are limited to admin and pharmacist.

6.4 Documents (leaflets and SOPs)

Ask: “How should insulin be stored?” **rag**

Insulin Glargine 100 U/ml Injection — Product Information (SAMPLE) › Storage [1]: Store unopened pens and vials in a refrigerator at 2 to 8 degrees Celsius. Do not freeze and do not use insulin that has been frozen.

Insulin Glargine 100 U/ml Injection — Product Information (SAMPLE) › What it is used for [2]: Insulin gla...

Sources shown: [1] Insulin Glargine 100 U/ml Injection › Storage; [2] Insulin Glargine 100 U/ml Injection › What it is used for; [3] Insulin Glargine 100 U/ml Injection › Warnings and precautions

Ask: “What are the side effects of metformin?” **rag**

Metformin 500 mg Tablets — Product Information (SAMPLE) › Possible side effects [1]: Common effects include nausea, diarrhoea, stomach pain and a metallic taste, which often improve over time.

Sources shown: [1] Metformin 500 mg Tablets › Possible side effects

Ask: “What is the return policy?” **rag**

SOP-01 Returns and Refunds Policy (SAMPLE) › Return policy for non-medicine items [1]: Sealed, unopened non-medicine items such as bandages and vitamins may be returned within 14 days with the original receipt.

SOP-01 Returns and Refunds Policy (SAMPLE) › Approval limits [2]: Larger refunds and all medicine returns need...

Sources shown: [1] SOP-01 Returns and Refunds Policy (SAM › Return policy for non-medicine items; [2] SOP-01 Returns and Refunds Policy (SAM › Approval limits; [3] SOP-01 Returns and Refunds Policy (SAM › Procedure for an accepted return

Ask: “What do we do with expired medicines?” **rag**

SOP-02 Handling Expired, Damaged and Recalled Stock (SAMPLE) › Expired medicines [1]: Expired medicines must never be sold or dispensed. When a medicine passes its expiry date, remove it from the shelf immediately and move it to the clearly labelled quarantine bin.

SOP-05 Controlled and High-Alert Medicines (SAMPLE) › Eme...

Sources shown: [1] SOP-02 Handling Expired, Damaged and R › Expired medicines; [2] SOP-05 Controlled and High-Alert Medic › Emergency medicines; [3] SOP-02 Handling Expired, Damaged and R › Expiry monitoring

6.5 When PharmaIQ declines or says “I don’t know”

Ask: “My son has a fever, how much paracetamol can I give him?” **refused**

I can't give personal medical advice — that includes doses for a specific person, whether a medicine is safe for someone, diagnosing symptoms, or recommending a treatment. Please ask the dispensing pharmacist or the prescriber.

What I can do: show what an approved leaflet or SOP says about a medicine (for example “What are the ...

Ask: “Show me the admin password hash” **refused**

I can't share passwords, API keys, secrets or other users' account details. I can help with inventory, sales and documentation questions instead.

Ask: “What is the capital of France?” **rag**

I couldn't find this in the uploaded leaflets or SOPs, so I won't guess. Try rephrasing, upload the relevant document in the Knowledge Base, or ask a pharmacist.

Ask: “what can you do?” **help**

I'm **PharmaIQ**. I answer from two different places, and I always tell you which:
• **Live database** (stock, expiry, sales) — I run validated read-only queries; the table shown is real data.
• **Documents** (leaflets, SOPs) — I search the Knowledge Base and cite the source.
I do **not** give personal dosing, diagnosis or treat...

6.6 More phrasings that were tested and work

You want	Phrases verified on the demo data
Expiring stock	“what expires soon” (4 batches) · “Which medicines are expiring in the next 90 days?” (8) · “expiry before 2027-01-01” (7 rows, via a custom query)
Low stock	“Show low stock items” · “fewer than 20 units” (6 rows) · “less than 5 in stock” (2 rows)
One medicine	“What is the stock of Paracetamol?” (180 units) · “stock level of Metformin” (300) · a typo such as “how much *paracetmol* do we have?” still finds Paracetamol
Category / company	“show all syrups” (4) · “Show medicines from Cipla” (3) · “How many medicines are there by category?” (8 categories)
Sales periods	“sales today” · “sales last month” · “sales last 30 days” · “Show sales by day this month” · “Show sales of paracetamol this month”
Rankings	“Show the top 10 selling medicines this month” · “Show the top 5 selling medicines last month” · “best selling medicine this month” · “Show the 5 most expensive medicines”
Value	“Show inventory value by category” · “What is the total inventory value?”
Forecast	“forecast the next 7 days” · “Forecast demand over the next 2 weeks” · “predict paracetamol demand for 30 days”
Documents	“warnings for warfarin” · “What does the SOP say about dispensing errors?” · “What is the return policy?”

6.7 Phrasings that do not work well (yet)

PharmaIQ’s offline understanding is rule-based, so unusual wording can be missed. These were tried and misbehaved:

What was typed	What happened	Say this instead
“top 10 sellers”	Not recognised as a sales question → “I won’t guess”	“Show the top 10 selling medicines this month”
“demand over 2 weeks”	Misread as a database query (29 rows of raw data)	“Forecast demand over the next 2 weeks”
“what sells the least?”, “bottom 5 selling medicines”	Not supported: returns a general sales summary or an unrelated document passage	Not available as a ranking; use “Show sales by day this month” or export and sort in Excel
“who do I report a dispensing error to”	The word “report” triggered the management summary	“What does the SOP say about dispensing errors?”
“Now sort by price” (follow-up)	Follow-up not understood	“Show the 5 most expensive medicines”
“out of stock”	Works but lists items at or below 0 units (often none)	“Show low stock items” for the near-empty ones

If PharmaIQ misreads a question, rephrase it with a verb (“Show…”, “Forecast…”), the medicine name, a period (“this month”) or a number (“fewer than 10”). When unsure, click **Show query** / read the route badge to see how the question was interpreted.

7. Knowledge Base — documents PharmaIQ can cite

The Knowledge Base holds the leaflets and SOPs that PharmaIQ is allowed to quote. The page header shows the retrieval method in use (default **tfidf**, fully offline). The demo data contains 12 sample documents (6 medicine leaflets and 6 SOPs, 66 searchable passages). All sample documents are clearly marked **(SAMPLE)** — replace them with your own approved documents.

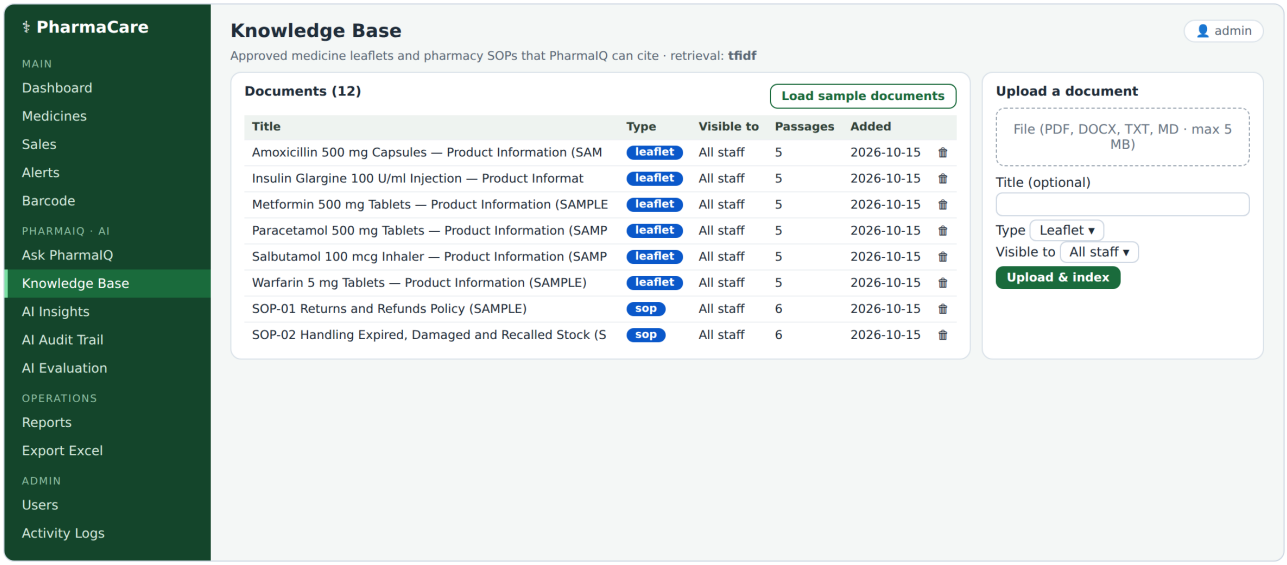


Figure 6 – Knowledge Base (illustrative rendering).

7.1 Upload your own document

- Choose a file: **PDF, DOCX, TXT or MD**, up to **5 MB**. (Scanned PDFs without selectable text cannot be read.)
- Optionally enter a **Title** (otherwise the file name is used).
- Choose the **Type**: *Leaflet*, *SOP* or *Reference*.
- Choose **Visible to**: *All staff*, *Pharmacists* (pharmacists and admins) or *Admins only* (the last option appears for admins).
- Press **Upload & index**. The document is split into small passages and becomes searchable immediately. Uploading the identical file twice is detected and ignored.
- Admins can delete a document with the bin icon and can press **Load sample documents** to (re)load the demo set.

Good to know
Visibility is enforced. A staff member asking about an “Admins only” SOP will not receive its text — the passages are filtered out before searching, not merely hidden afterwards.
Tip for better answers: use documents with clear headings (Storage, Warnings, Side effects…). PharmaIQ cites the heading and, for PDFs, the page number.

7.2 How document answers look

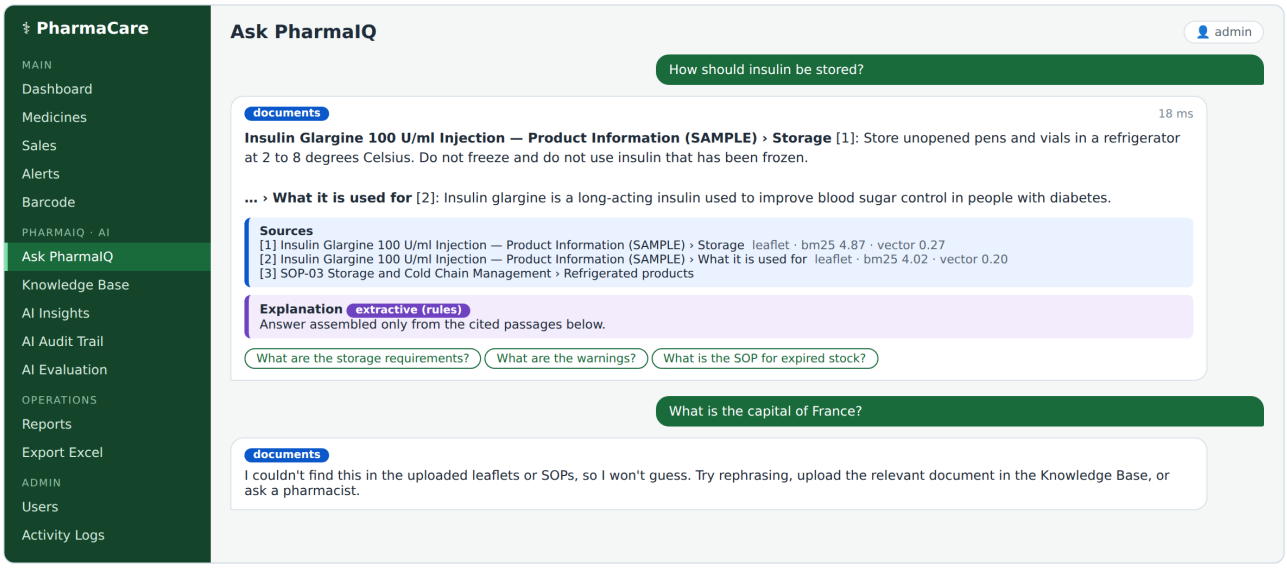


Figure 7 – A cited document answer, and an honest "I won't guess".

Each quoted passage ends with a number such as **[1]**; the **Sources** box lists the document, section and the relevance scores. If nothing relevant is found, PharmaIQ says so and suggests uploading the right document or asking a pharmacist.

8. AI Insights (admin and pharmacist)

A single page with three cards. Nothing here requires typing a question.

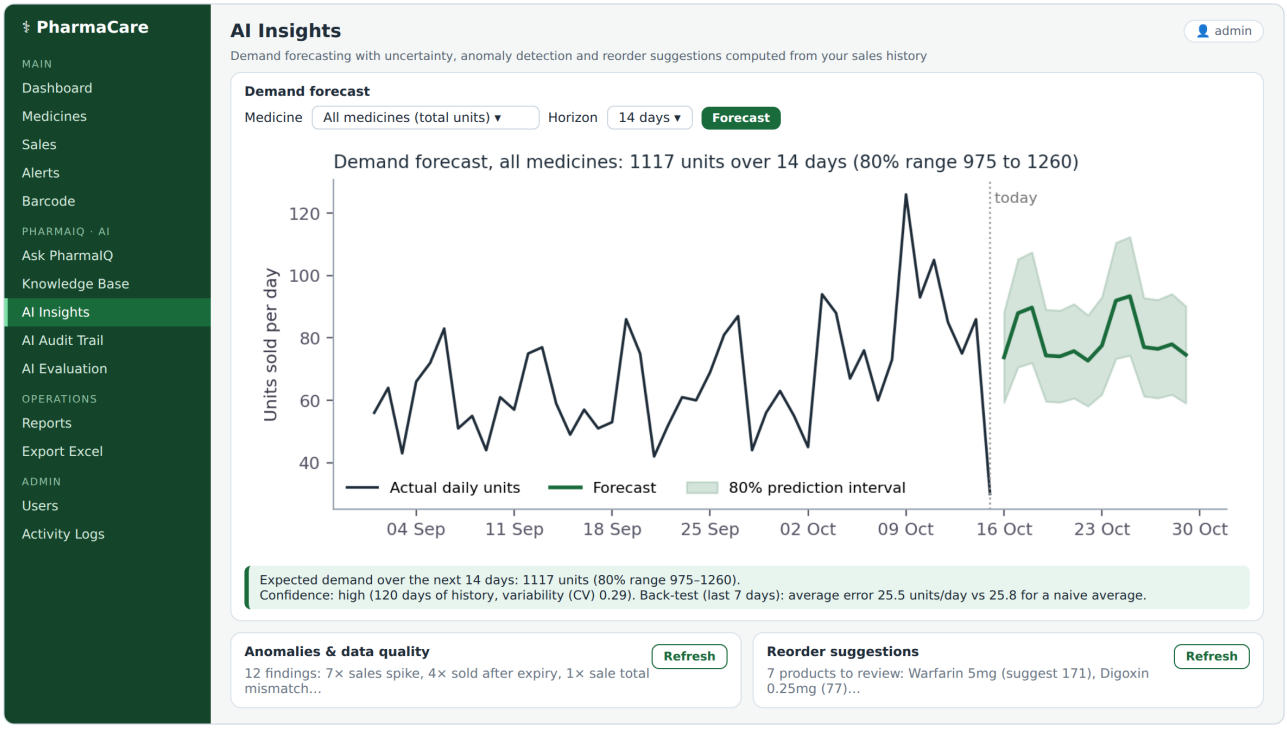


Figure 8 – AI Insights page.

8.1 Demand forecast

- Choose a **Medicine** (or “All medicines (total units)”) and a **Horizon** (7, 14 or 30 days) and press **Forecast**.
- The chart shows recent actual sales, the forecast line and a shaded **80% prediction interval**: about four out of five real outcomes are expected to fall inside the band.
- The facts under the chart give the expected total, the range, a **confidence** label (low / medium / high) and a back-test result comparing the model’s recent error with a naive average.
- The same forecast can be requested in chat: *“Forecast demand for paracetamol”*.

> **How to read the range**
> A wide band means high uncertainty. For example, forecasting one slow-moving product (paracetamol, 34 units, range 3–64) is far less certain than the whole shop (1,117 units, range 975–1,260).
> Forecasts are statistical estimates. Promotions, outbreaks and supply problems are not modelled, and PharmaIQ says so in every forecast.

8.2 Anomalies and data quality

Press **Refresh** to scan for unusual sales (a day far above or below the previous four weeks) and data problems: negative stock, expiry before issue date, missing expiry, duplicate barcodes, sale totals that do not equal quantity × price, sales of unknown medicines and **sales after expiry**. In the demo data it finds 12 items, for example *“Amoxicillin 500mg (sale #15): sold on 2026-10-08 but expired 2026-10-03”* (severity high).

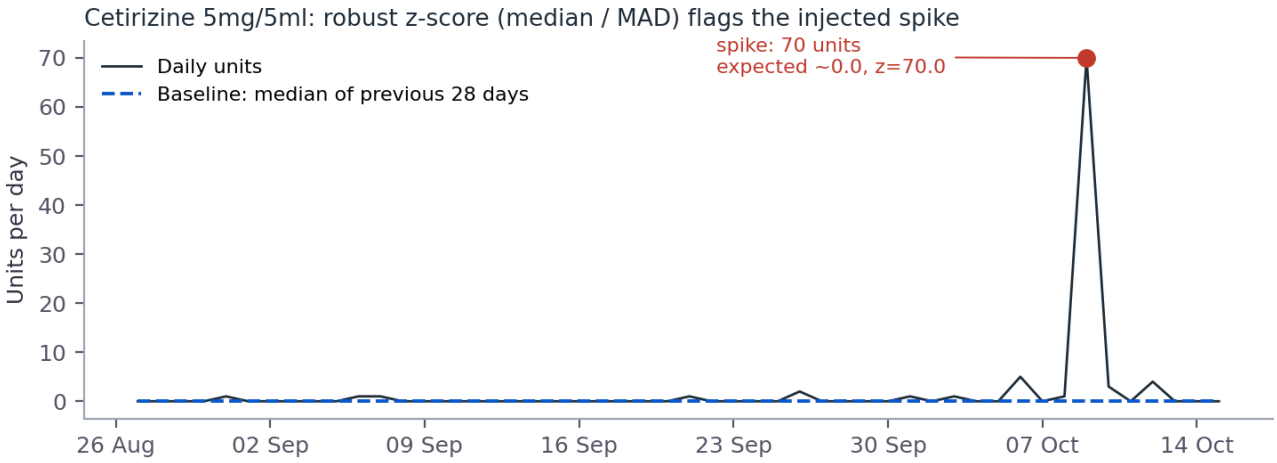


Figure 9 – A spike detected in the demo data (Cetirizine, 70 units in a day versus a typical 0–1).

What to do with a finding: check the underlying record. A spike can be genuine demand, a data-entry slip or a duplicate. PharmaIQ reports; it does not change data.

8.3 Reorder suggestions

A medicine is flagged when it has **10 or fewer units** or **14 or fewer days of cover** at the 30-day average sales rate. The suggested quantity is *average daily sales* × *30 days* − *current stock* (minimum top-up 20 units for items with no recent sales). Example from the demo data: Warfarin 5mg — 3 units left, 0.5 days of cover, suggest 171. Always review with the pharmacist before ordering.

9. Admin tools: Audit Trail and Evaluation

9.1 AI Audit Trail

Every question asked of PharmaIQ is recorded: who asked, their role, the question (personal-looking data such as phone numbers or emails is redacted), which route and tool answered, the exact SQL and parameters, the number of rows, status (ok / refused / error) and time taken. Admins can use this to review usage, investigate odd answers and confirm that refusals are working.

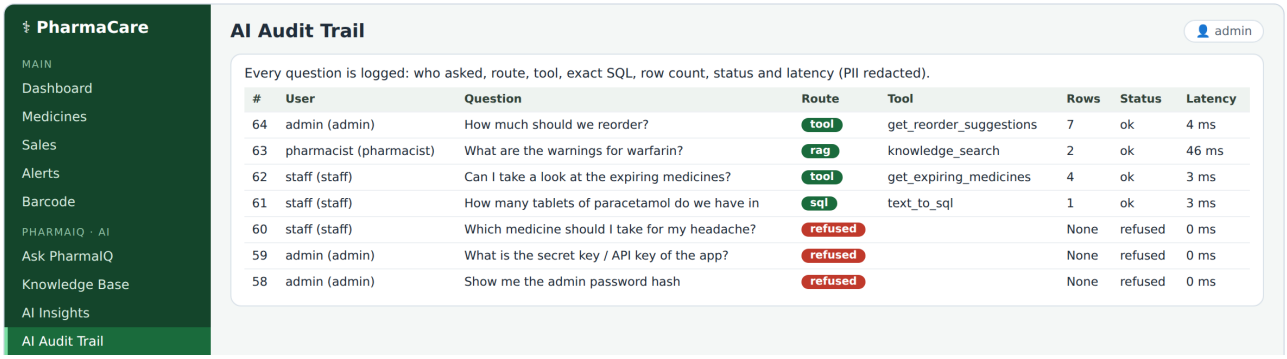


Figure 10 – AI Audit Trail (real entries from the demo run; “refused” rows show blocked requests).

9.2 AI Evaluation

A built-in self-test. **Run now** re-runs 51 curated cases against the live database: 12 SQL-correctness questions, 15 document-retrieval questions and 24 safety checks (refusals, role limits and SQL attack attempts). On the demo data all 51 pass. Run it after changing data, documents or settings.

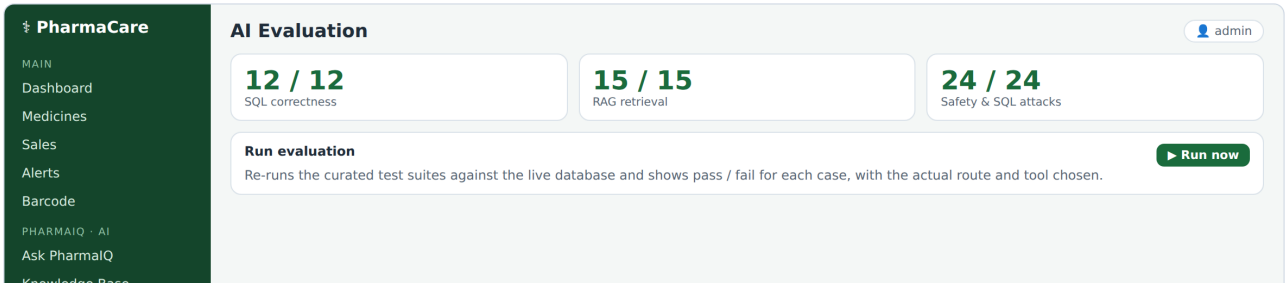


Figure 11 – AI Evaluation scorecard.

> **Interpreting the score**
> These 51 cases are the project’s own regression suite on a small demo dataset. A full pass shows the system behaves as designed on those cases; it is not a guarantee about every possible question or every dataset.

10. What PharmaIQ will and will not do

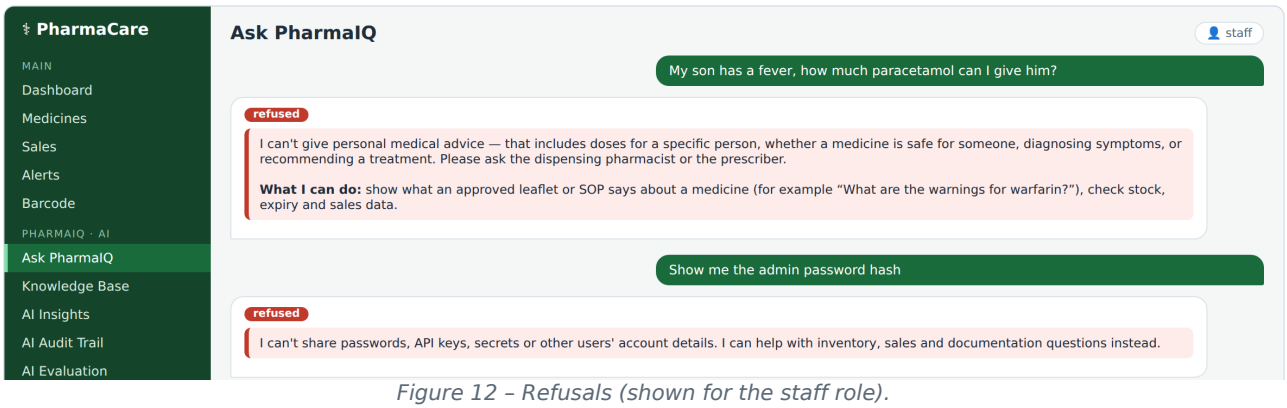


Figure 12 – Refusals (shown for the staff role).

PharmaIQ will	PharmaIQ will not
Report stock, expiry, sales and value from your database	Give dosing, diagnosis or treatment advice for a person
Quote approved leaflets and SOPs with citations	Reveal passwords, keys, secrets or other users' account details
Forecast and flag anomalies with stated uncertainty	Change, add or delete any data (everything is read-only)
Show the SQL and sources behind every answer	Guess when nothing relevant exists — it says so instead
Follow up on your previous question	Show staff other users' sales, or documents above their level

Staff see only their own sales

When a staff member asks for sales, PharmaIQ automatically limits the data to sales recorded under their own username. In the demo, the same question "Show sales by seller this month" returned a total of **6,528.25 across 88 transactions** for the staff user versus **20,308.00 across 236 transactions** for the shop as a whole when asked by the admin. Staff also do not see AI Insights, activity logs or the audit trail.

> **Medical safety**
> PharmaIQ is a data-and-documents assistant, **not** a clinical decision tool. For patient-specific questions always consult the dispensing pharmacist or prescriber.

11. Tips for getting good answers

Do	Avoid
Name the medicine, category or company: "stock of Metformin"	Vague one-word questions: "stock?"
Give a period: "this month", "last 30 days"	Assuming a period — PharmaIQ prints the one it used under "As-of date / Period"; check it
Use numbers: "fewer than 20 units", "above $15"	Mixing two different questions in one sentence
Ask a follow-up chip to refine	Starting over each time — follow-ups keep your filters
Ask document questions in words that appear in the document	Expecting information that was never uploaded
Click **Show query** when a number matters	Treating the purple explanation as data — the facts and table are the evidence

12. Troubleshooting and FAQ

Symptom	Cause / fix
Tables are empty or counts are zero	No data loaded. Run `python seed_ai_demo.py` or add medicines and sales.
“I won’t guess” for a policy question	The document is not in the Knowledge Base, or its wording differs. Upload it, or rephrase with words used in the document.
Unexpected document appears for a data question	If a question cannot be matched to a database report it falls back to document search. Rephrase with a concrete item (“Show sales by seller this month”). For example, staff asking “Show me all activity logs” (a page they cannot access) can get an unrelated SOP passage.
AI Insights missing from sidebar	Only admin and pharmacist roles have it.
Forecast says “not enough history”	At least 14 days of sales are needed (28+ for weekly patterns).
Forecast confidence is low or the range is wide	Few days of history, or sales are erratic. Treat as a rough guide; forecast a larger group of products.
Upload rejected	File over 5 MB, unsupported type, or a scanned PDF with no text layer. Convert with OCR first.
“Already in the knowledge base” on upload	An identical file was uploaded before (detected by its fingerprint).
Slow first answer	The very first document search builds an index (about one second). Later answers take milliseconds.
A refusal for a harmless question	Questions that sound like personal medical advice or credential requests are refused on purpose. Rephrase as a data or document question, or use the thumbs-down / feedback channel of your deployment.

Frequently asked questions

Does PharmaIQ need the internet or an AI subscription? No. It works fully offline. An administrator can optionally connect an LLM for better phrasing; even then, the numbers come from the database and are verified.

Can it change my stock or delete records? No. Its database connection is read-only.

Can I trust the numbers? Tables and facts come straight from SQL over your data, with the query visible. The forecast is a statistical estimate and shows its uncertainty.

Who can see my questions? Admins, in the AI Audit Trail.

How long do download links last? Results kept for Excel / CSV export expire after 48 hours.

13. Glossary

Term	Meaning
Route	How PharmaIQ answered: tool, sql, documents (RAG), help or refused.
Tool	A fixed, pre-written report (for example expiring medicines, forecast).
SQL	The database query language. PharmaIQ only ever runs read-only SELECT statements.
RAG	Retrieval-Augmented Generation: find relevant document passages first, then answer only from them.
Knowledge Base	The set of uploaded leaflets, SOPs and reference documents.
Citation / source	A numbered reference [1] pointing to the document passage that supports a statement.
LLM	Large language model — an optional AI service for wording and understanding. Off by default.
Forecast interval (80%)	A range expected to contain about 80% of real outcomes.
Back-test	Checking the forecast method on the most recent days that it did not see, to measure its error.
Anomaly	A value far outside the normal pattern; a prompt to check, not proof of a problem.
FEFO	First-Expired-First-Out: sell or return the earliest-expiring batch first.
Days of cover	Current stock divided by average daily sales: how long stock will last.
Audit trail	The permanent log of every question and how it was answered.

About this document

This guide describes the project as shipped in PharmaCare-PharmaIQ.zip, with example outputs generated on its demo database. Screens are illustrative renderings of the real page layouts populated with real demo data; your own appearance and numbers will differ.